

Reading DNA to Predict Regulatory Activity and the Effect of Single-Base Changes

A sequence-to-function deep learning pipeline on developing human cortex MPRA data
Data: Song, Pollard & Ahituv *et al.* massively parallel reporter assay (Science, adh0559). Model, variant-effect, interpretation, design, and multi-cell analyses.

Executive summary

This document summarizes the complete pipeline and results. The final cortex-specific model achieves held-out Spearman **0.614 / 0.595** (primary / organoid) with is-silencer AUROC **0.80**, using multi-task activity + is-active + silencer heads and one non-B DNA channel.

We trained a convolutional model that reads a raw DNA sequence and predicts its regulatory activity in a developing-cortex MPRA, then used it to ask what a single-base change does. On completely held-out chromosomes the model reaches Spearman **0.60–0.61** (primary cortex / organoid), roughly double the k-mer baseline and clearly above the previous CNN, and separates active from silencer elements at AUROC ~0.80. Built on top of it: an ensemble variant-effect model that flags functionally active variants (AUROC 0.63), saturation-mutagenesis maps that rank every base of a candidate element, motif interpretation matched to JASPAR transcription factors, working synthetic-enhancer design, and a multi-cell-type transfer test. Every result below is on held-out data, and the conclusions include the honest negatives.

1. Data provenance — verified

Before modelling we verified the input against the original Science supplement. The activity table is **byte-identical** to Supplement S1 (46,370 primary + 43,902 organoid inserts; 100% of insert names matched; is-active labels 100% identical). The variant library is byte-identical to Supplement S2 (31,406 rows; measured log-fold-changes agree to floating-point precision, 10^{-16}). The model is trained on genuine, unmodified Pollard/Ahituv data — not synthetic data.

2. Activity model — reading DNA to predict activity

Architecture (“RegNetDNA”): a wide motif-detecting first convolution, a tower of residual dilated convolution blocks with squeeze-excite attention, attention pooling, and reverse-complement equivariance, trained as a 3-seed ensemble with multi-task heads. Held-out chromosomes (test = chr7+chr17, val = chr8+chr9).

Held-out test	k-mer ridge	previous CNN	RegNetDNA
Spearman — Primary	0.345	0.558	0.603
Spearman — Organoid	0.322	0.528	0.585
R2 (variance expl.) — P / O	0.08 / 0.06	0.28 / 0.25	0.37 / 0.34
is-active AUROC — P / O	—	0.78 / 0.77	0.805 / 0.797

3. Multi-task extensions: silencer head and non-B DNA channels

Adding a silencer head (Primary labels) let the shared encoder learn **repression** grammar, not just activation: is-silencer AUROC ~**0.80** on held-out data. We then ablated non-B DNA structural channels. One channel of G-quadruplex propensity (G4Hunter) adds a small but consistent lift; i-motif is numerically identical to G4 (it is the same signed C/G signal) and therefore redundant; R-loop (GC-skew) adds nothing on 270 bp oligos. Verdict: **one non-B channel is enough**.

Model (Primary / Organoid Spearman)	Primary	Organoid	is-silencer AUROC
Multi-task, no non-B	0.597	0.583	0.795
+ G4	0.608	0.588	0.794
+ i-motif ($\equiv$ G4)	0.614	0.595	0.799
+ R-loop	0.606	0.585	0.793
+ all three	0.612	0.596	0.804

Best cortex model: multi-task + one non-B channel + silencer head — Spearman **0.614 / 0.595**, is-active AUROC 0.81, is-silencer AUROC 0.80, in a single model.

4. Single-base variant effects

Honest framing: of 31,406 variants only ~4.4% are statistically significant and the median effect is tiny, so scoring every variant lands near zero *by construction*. The signal lives in the significant subset. An ensemble Siamese model (warm-started from the activity encoder, confidence-weighted) was trained and reported on the targets that carry signal:

Held-out metric	Primary: single → ensemble	Organoid: single → ensemble
Functional AUROC (flags real variants)	0.549 → 0.632	0.581 → 0.602
Spearman, significant	0.148 → 0.347	0.137 → 0.248
Sign accuracy, significant	0.564 → 0.590	0.588 → 0.571

The ensemble + confidence weighting roughly doubled the significant-subset correlation and lifted functional AUROC to 0.63 — a real, modest signal concentrated exactly where the biology is. Per-variant single-base prediction is intrinsically hard on this noisy GWAS/eQTL library; these are the honest numbers.

5. What does a single base do — saturation mutagenesis

For the top prioritised candidates we scored every position × every base (in-silico saturation mutagenesis) and ranked each real variant against the full mutational landscape of its element. Of 30 candidates, ~8 are **high-impact** (the variant sits on a base the model strongly weights): e.g. rs11080641 (SMARCA1, 98th percentile), rs79804582 (100th), rs113114378 (ZFP28, 99th). For others (e.g. the SOX2-motif rs2883420, 13th percentile) the element is active but the SNP itself is a low-impact base — an informative negative.

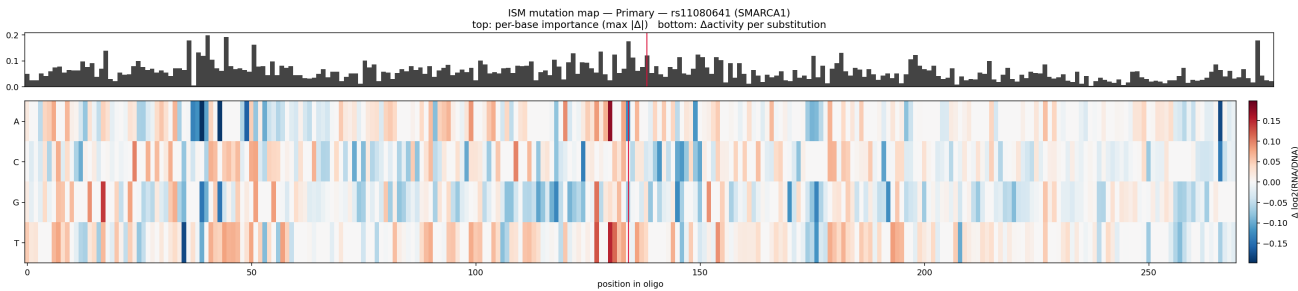


Figure 1. Saturation-mutagenesis map for rs11080641 (SMARCA1). Top: per-base importance; bottom: Δ activity per substitution (blue = lowers activity; red line = the assayed variant position (here a high-impact base)).

6. Motif interpretation matched to JASPAR

The model's first-layer filters were extracted as motifs, validated by enrichment (79/192 filters separate active vs inactive oligos at $p < 10^{-3}$), and matched to JASPAR2024 transcription factors. The pattern is biologically coherent and reproduces across both assays: **activating** motifs match neuronal TFs (PHOX2A/PHOX2B, MEF2/FLI1); **repressive** motifs match zinc-finger repressors (ZNF213/418, BCL6), E2F4, IRF9, and the MAF/Zic families. Official MEME-TOMTOM (memelite, complete-score algorithm) confirms this with hard statistics: **33/192 learned motifs match a JASPAR TF at $q < 0.05$** , led by **CTCF** (the GC-rich architectural motifs), **GATA1::TAL1**, **MAF/MAFA**, **Pou5f1::Sox2**, and nuclear receptors (PPARA::RXRA) — p-values $\sim 10^{-6}$.



Figure 2. Learned motifs (left of each pair) matched to their best JASPAR transcription factor (right). Activating filters map to neuronal TFs; repressive filters to zinc-finger/BCL6/E2F4 repressors. Top matches shown; full statistics in `jaspar_match_*.csv` and official TOMTOM q-values in `tomtom_Primary.csv`.

7. Synthetic enhancer design

Running the model in reverse (in-silico directed evolution from random DNA) reliably designs sequences it predicts as top-1% enhancers: five designs converged to a mean predicted activity of 1.96 vs the real library's 99th percentile of 1.88. Caveat: these are model predictions and optimising against a model can exploit its blind spots — they are strong hypotheses for a design-validation MPRA, not proven enhancers.

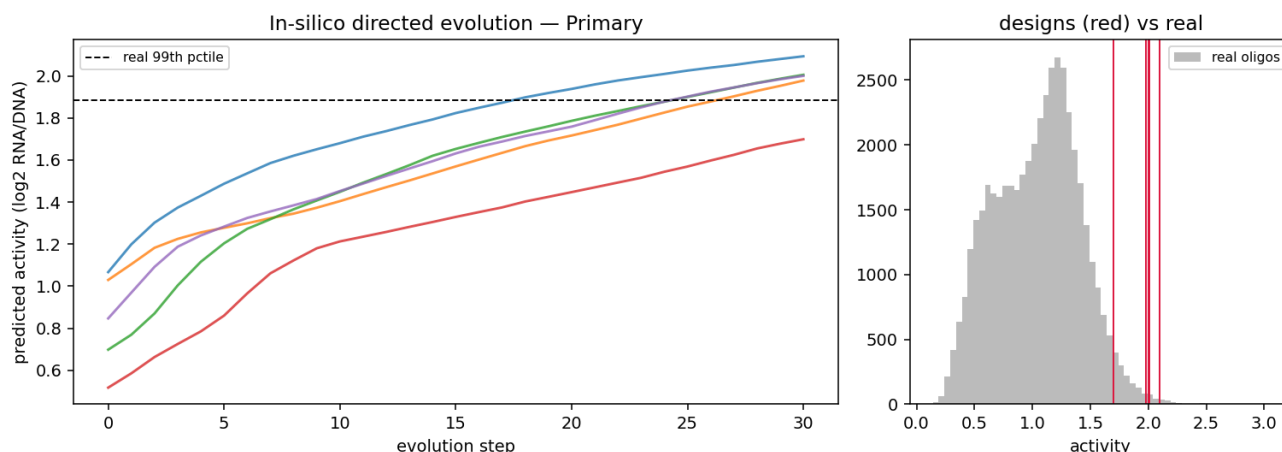


Figure 3. Directed evolution. Left: five sequences climb past the real 99th-percentile activity (dashed). Right: designs (red) land in the far-right tail of the real activity distribution (grey).

8. Multi-cell-type transfer (WTC11) — a clean negative

We tested whether the Agarwal/LegNet WTC11 lentiMPRA (46,185 elements) could improve the cortex model, two ways: as a third equal-weight task head, and as a pretrain→fine-tune transfer. The integration itself works — the model predicts WTC11 activity at Spearman 0.603 through the shared trunk — but neither route helped cortex; both *hurt* it:

Cortex held-out Spearman	Primary	Organoid
Cortex-only (best)	0.614	0.595
Multi-task + WTC11 (equal heads)	0.603	0.585
Pretrain WTC11 → fine-tune cortex	0.558	0.537

WTC11 is a large, *non-neural* library; as an equal head it competes for encoder capacity, and as a pretraining set it drops cortex into a worse optimum the fine-tune cannot escape. Verdict: for this cortex MPRA, off-target cell-type data does not help — **the cortex-specific signal is best captured by the cortex data itself**. 'More data from other cell types' is not the lever here.

9. Honest limitations and next steps

- Activity prediction is near this assay's ceiling — a larger 5-seed model gave no gain, so the data, not the architecture, is the limit.
- Single-base variant effect is modest (functional AUROC ~0.6) because the library is dominated by small/noisy effects; the exact oligo-design coordinates or a larger MPRA corpus would help most.
- Cross-cell-type augmentation was tested and does not help this cortex target (both multi-task and pretrain→fine-tune underperformed cortex-only) — the ceiling is set by the cortex data itself.
- Motif TFs were confirmed with official MEME-TOMTOM statistics (33 significant at $q < 0.05$).
- Highest-value remaining work: the exact oligo-design coordinates from the paper's supplement (to sharpen single-base variant effects), and wet-lab validation of the top saturation-mutagenesis candidates and the synthetic designs.

Appendix — key files

models/activity_mt_i.pt (best cortex model) · models/variant_ens.pt (variant effects) · results/variant_priority.csv, satmut_summary_Primary.csv, jaspar_match_*.csv, designs_Primary.fasta, learned_motifs_Primary.meme. Scripts under src/: models_best, train_activity, train_variant_ens, ism, satmut, motifs, jaspar_match, motif_enrich, design, prepare_promoter, train_activity_mt.